



10464 - Echoes of the First Quasars

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
J0252-0503				
	1	J0252 MPT P1	NIRSpec MultiObject Spectroscopy	(3) J0252_all_targets_cy5_weights
	2	J0252 MPT P2	NIRSpec MultiObject Spectroscopy	(4) J0252_all_targets_cy5_nw
J1007+2115				
	3	J1007 MPT P1	NIRSpec MultiObject Spectroscopy	(2) J1007_all_targets_cy5
	4	J1007 MPT P2	NIRSpec MultiObject Spectroscopy	(2) J1007_all_targets_cy5

ABSTRACT

The epoch of reionization represents the triumph of light over matter in the first billion years of cosmic time, the last global phase transition of intergalactic gas from a cold, neutral state to the warm ionized medium that pervades the Universe today. The best constraints on the history of the reionization process come from Lyman-alpha absorption spectroscopy of the most distant quasars, powered by rapid accretion onto the earliest supermassive black holes. Past analyses have relied on a one-dimensional down-the-barrel view of the ionized and neutral gas along the line of sight. The tremendous sensitivity of JWST imaging and spectroscopy enable a leap into three dimensions via the ionizing "light echo" imprinted by quasars in the Lyman-alpha forest of background galaxies, as recently demonstrated by the JWST MASQUERADE program. We propose deep 12-29 hr NIRSpec G140M integrations on 19 spectroscopically-confirmed galaxies in the fields of two $z > 7$ quasars to constrain their local reionization topology, their UV-luminous lifetime, and obscuration geometry. In the process, we will also characterize the stars and gas within a UV-bright companion galaxy less than 10 kpc from one of the quasars, and determine the nature of the rest-UV light of a luminous $z=7.27$ little red dot in the foreground of the other.

OBSERVING DESCRIPTION

We will observe the fields towards the $z=7.00$ quasar J0252-0503 (J0252) and the $z=7.52$ quasar J1007+2115 (J1007) with deep NIRSpec/MSA G140M/F070LP spectroscopy, targeting galaxies behind and in the nearby environments of the two quasars.

MSA configurations have been designed to observe our targets with specific aperture position angles to maximize the number of sources per pointing. The exposures will be taken with the NRSIRS2 readout mode, 20 groups per integration, and 2 integrations per 3-slitlet dither position.

1. J1007 -- Two MSA masks have been designed with APA = 61.71 degrees and APA = 254.38. degrees to observe 8 primary objects (3 objects in common). Each mask will be observed for 5 sets of 3-slitlet dithers, totalling 12.2 hr of exposure time.
2. J0252 -- Two MSA masks have been designed with APA = 28.0 degrees and APA = 194.85 degrees to observe 12 primary objects (5 objects in common). Each mask will be observed for 6 sets of 3-slitlet dithers, totalling 14.4 hr of exposure time.

Additional slitlets will be opened on additional photometric candidate galaxies during Phase 2 observation preparations.

During the long NIRSpec integrations, we will obtain NIRCам parallel imaging of outlying fields to select potential future light echo targets, both behind and at the same redshift as the two quasars. Each 3-dither MSA sequence will have a corresponding set of 7-group 2-integration DEEP8

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exposures (2.3 hr per sequence) chosen to roughly match the NIRSpec exposure time while keeping the data volume below the limit. For the J1007 field, we will obtain 3 F090W, 1 F115W, and 1 F150W sequences in the SW channel, and 1 F277W, 1 F356W, 1 F444W, 1 F410M, and 1 F430M sequences in the LW channel. For the longer MSA exposure in the J0252 field, we add one additional sequence of F090W and F430M.

Proposal 10464 - Targets - Echoes of the First Quasars

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	J1007_all_targets_cy5	RA: 10 08 0.3300 (152.0013750d) Dec: +21 15 38.71 (21.26075d) Equinox: J2000		
	Comments: Description=[]				
	(3)	J0252_all_targets_cy5_weight s	RA: 02 52 16.5624 (43.0690100d) Dec: -05 03 30.42 (-5.05845d) Equinox: J2000		
	Comments: Description=[]				
	(4)	J0252_all_targets_cy5_nw	RA: 02 52 16.5624 (43.0690100d) Dec: -05 03 30.42 (-5.05845d) Equinox: J2000		
	Comments: Description=[]				

Proposal 10464 - Observation 1 - Echoes of the First Quasars

Observation	Proposal 10464, Observation 1: J0252 MPT P1											Fri Mar 13 19:07:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	J0252_all_targets_cy5_weights	RA: 02 52 16.5624 (43.0690100d) Dec: -05 03 30.42 (-5.05845d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCам Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: J0252_all_targets_cy5_weights (13 sources)											
	Filler Candidate List: null											
Reference Stars	Spectral Overlap Map: jwst-nirspec-g140m											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	NONE										

Proposal 10464 - Observation 1 - Echoes of the First Quasars

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.079989166666 67 Degrees - 5.049030555555 32 Degrees	194.84902897761 73			3	6	8840.867
	2	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.079989166666 67 Degrees - 5.049030555555 32 Degrees	194.84902897761 73			3	6	8840.867
	3	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.079989166666 67 Degrees - 5.049030555555 32 Degrees	194.84902897761 73			3	6	8840.867
	4	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.079989166666 67 Degrees - 5.049030555555 32 Degrees	194.84902897761 73			3	6	8840.867
	5	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.079989166666 67 Degrees - 5.049030555555 32 Degrees	194.84902897761 73			3	6	8840.867
	6	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.079989166666 67 Degrees - 5.049030555555 32 Degrees	194.84902897761 73			3	6	8840.867
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	DEEP8	7	2	6	3	8278.05		
	2	F090W	F356W	DEEP8	7	2	6	3	8278.05		
	3	F090W	F444W	DEEP8	7	2	6	3	8278.05		
	4	F090W	F410M	DEEP8	7	2	6	3	8278.05		
	5	F115W	F430M	DEEP8	7	2	6	3	8278.05		
	6	F150W	F430M	DEEP8	7	2	6	3	8278.05		
Special Requirements	Aperture PA Range 194.85 to 194.85 Degrees (V3 56.2754303 to 56.2754303) No Parallel Attachments MSA Planned Aperture PA 194.8500 to 194.8500 Degrees (V3 56.2754303 to 56.2754303)										

Proposal 10464 - Observation 2 - Echoes of the First Quasars

Observation	Proposal 10464, Observation 2: J0252 MPT P2											Fri Mar 13 19:07:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	J0252_all_targets_cy5_nw	RA: 02 52 16.5624 (43.0690100d) Dec: -05 03 30.42 (-5.05845d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCам Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: J0252_all_targets_cy5_nw (13 sources)											
	Filler Candidate List: null											
Reference Stars	Spectral Overlap Map: jwst-nirspec-g140m											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	NONE										

Proposal 10464 - Observation 2 - Echoes of the First Quasars

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.067773208333 335 Degrees - 5.057408333333 42 Degrees	28.000111987991 68			3	6	8840.867
	2	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.067773208333 335 Degrees - 5.057408333333 42 Degrees	28.000111987991 68			3	6	8840.867
	3	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.067773208333 335 Degrees - 5.057408333333 42 Degrees	28.000111987991 68			3	6	8840.867
	4	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.067773208333 335 Degrees - 5.057408333333 42 Degrees	28.000111987991 68			3	6	8840.867
	5	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.067773208333 335 Degrees - 5.057408333333 42 Degrees	28.000111987991 68			3	6	8840.867
	6	1 (G140M/F070LP)	c1	3 Shutter Slitlet	43.067773208333 335 Degrees - 5.057408333333 42 Degrees	28.000111987991 68			3	6	8840.867
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	DEEP8	7	2	6	3	8278.05		
	2	F090W	F356W	DEEP8	7	2	6	3	8278.05		
	3	F090W	F444W	DEEP8	7	2	6	3	8278.05		
	4	F090W	F410M	DEEP8	7	2	6	3	8278.05		
	5	F115W	F430M	DEEP8	7	2	6	3	8278.05		
	6	F150W	F430M	DEEP8	7	2	6	3	8278.05		
Special Requirements	Aperture PA Range 28.0 to 28.0 Degrees (V3 249.42543030000002 to 249.42543030000002) No Parallel Attachments MSA Planned Aperture PA 28.0000 to 28.0000 Degrees (V3 249.42543030000002 to 249.42543030000002)										

Proposal 10464 - Observation 3 - Echoes of the First Quasars

Observation	Proposal 10464, Observation 3: J1007 MPT P1											Fri Mar 13 19:07:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	J1007_all_targets_cy5	RA: 10 08 0.3300 (152.0013750d) Dec: +21 15 38.71 (21.26075d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCам Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: J1007_primary (8 sources)											
	Filler Candidate List: J1007_all_targets_cy5 (18 sources)											
Reference Stars	Spectral Overlap Map: jwst-nirspec-g140m											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	NONE										

Proposal 10464 - Observation 3 - Echoes of the First Quasars

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.975530000000 002 Degrees 21.277358333333 332 Degrees	61.702769602509 28			3	6	8840.867
	2	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.975530000000 002 Degrees 21.277358333333 332 Degrees	61.702769602509 28			3	6	8840.867
	3	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.975530000000 002 Degrees 21.277358333333 332 Degrees	61.702769602509 28			3	6	8840.867
	4	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.975530000000 002 Degrees 21.277358333333 332 Degrees	61.702769602509 28			3	6	8840.867
	5	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.975530000000 002 Degrees 21.277358333333 332 Degrees	61.702769602509 28			3	6	8840.867
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	DEEP8	7	2	6	3	8278.05		
	2	F090W	F356W	DEEP8	7	2	6	3	8278.05		
	3	F090W	F444W	DEEP8	7	2	6	3	8278.05		
	4	F115W	F410M	DEEP8	7	2	6	3	8278.05		
	5	F150W	F430M	DEEP8	7	2	6	3	8278.05		
Special Requirements	Aperture PA Range 61.7121 to 61.7121 Degrees (V3 283.1375303 to 283.1375303) No Parallel Attachments MSA Planned Aperture PA 61.7121 to 61.7121 Degrees (V3 283.1375503 to 283.1375503)										

Proposal 10464 - Observation 4 - Echoes of the First Quasars

Observation	Proposal 10464, Observation 4: J1007 MPT P2											Fri Mar 13 19:07:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCам Imaging											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	J1007_all_targets_cy5	RA: 10 08 0.3300 (152.0013750d) Dec: +21 15 38.71 (21.26075d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCам Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: J1007_primary (8 sources)											
	Filler Candidate List: J1007_all_targets_cy5 (18 sources)											
Reference Stars	Spectral Overlap Map: jwst-nirspec-g140m											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	NONE										

Proposal 10464 - Observation 4 - Echoes of the First Quasars

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.98973875000 001 Degrees 21.248741666666 668 Degrees	254.37964839142 037			3	6	8840.867
	2	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.98973875000 001 Degrees 21.248741666666 668 Degrees	254.37964839142 037			3	6	8840.867
	3	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.98973875000 001 Degrees 21.248741666666 668 Degrees	254.37964839142 037			3	6	8840.867
	4	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.98973875000 001 Degrees 21.248741666666 668 Degrees	254.37964839142 037			3	6	8840.867
	5	1 (G140M/F070LP)	c1	3 Shutter Slitlet	151.98973875000 001 Degrees 21.248741666666 668 Degrees	254.37964839142 037			3	6	8840.867
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	DEEP8	7	2	6	3	8278.05		
	2	F090W	F356W	DEEP8	7	2	6	3	8278.05		
	3	F090W	F444W	DEEP8	7	2	6	3	8278.05		
	4	F115W	F410M	DEEP8	7	2	6	3	8278.05		
	5	F150W	F430M	DEEP8	7	2	6	3	8278.05		
Special Requirements	Aperture PA Range 254.3838 to 254.3838 Degrees (V3 115.8092303 to 115.8092303) No Parallel Attachments MSA Planned Aperture PA 254.3838 to 254.3838 Degrees (V3 115.8092653 to 115.8092653)										